SIMATS ENGINEERING

CO1-ASSESSMENT TOOL 2-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING COURSE **CODE: ITA0406**

COURSE OUTCOME COVERED

CO: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.

(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

1. Variables and Data Types Analysis (BL4 – Analyze)

Design pseudocode to simulate basic variable handling in R programming:

● Declare variables of different data types (numeric, character, logical)

● Perform arithmetic and logical operations

● Display the results

Pseudocode:

```
BEGIN
   DECLARE num1 AS NUMERIC
   DECLARE num2 AS NUMERIC
   DECLARE name AS CHARACTER
   DECLARE status AS LOGICAL
   SET num1 = 25
   SET num2 = 10
   SET name = "R Programming"
   SET status = TRUE
```

ADD num1 AND num2

STORE result in sum

SUBTRACT num2 FROM num1

STORE result in difference

MULTIPLY num1 AND num2

STORE result in product

DIVIDE num1 BY num2

STORE result in quotient

CHECK whether num1 > num2

STORE result in comparison

DISPLAY num1

DISPLAY num2

DISPLAY name

DISPLAY status

DISPLAY sum

DISPLAY difference

DISPLAY product

DISPLAY quotient

DISPLAY comparison

END

2. Data Structures Implementation (BL4 – Analyze)

Write pseudocode to demonstrate advanced data structures in R:

● Create and initialize a vector, list, and matrix

● Perform operations such as indexing, updating, and aggregation

● Compare the structures based on usage

Pseudocode:

```
BEGIN
    CREATE vector marks = [80, 70, 90, 60, 75]
    DISPLAY marks
    ACCESS the first element of marks
    UPDATE the second element to 85
    CALCULATE total of all elements in marks
    DISPLAY total
    CREATE list student containing:
        Name = "Ravi"
        Age = 20
        Mark = 85
    DISPLAY student list
    ACCESS student name
    UPDATE student mark to 90

    CREATE matrix M with 2 rows and 2 columns
    DISPLAY matrix M
    ACCESS an element from matrix M
    UPDATE the selected element
    CALCULATE sum of matrix elements
    DISPLAY sum
```

COMPARE structures:

Vector → stores similar type of data

List → stores different types of data

Matrix → stores data in rows and columns

END

3. Matrix and Array Operations (BL4 – Analyze)

Design pseudocode to perform operations on matrices and arrays:

● Create a matrix and a 3D array

● Perform addition, multiplication (matrix), and element-wise operations

● Display the results and analyze differences

Pseudocode:

BEGIN

CREATE matrix A with 2 rows and 2 columns

CREATE matrix B with 2 rows and 2 columns

DISPLAY A

DISPLAY B

ADD matrix A and matrix B

DISPLAY the result

MULTIPLY matrix A and B using matrix multiplication

DISPLAY the result

CREATE a 3D array with suitable dimensions

DISPLAY the 3D array

PERFORM element-wise addition on the array

DISPLAY the result

COMPARE matrix and array operations

DISPLAY:

Matrix → two-dimensional structure

Array → can have multiple dimensions

Matrix multiplication follows matrix rules

Element-wise operations work on corresponding elements

END

4. Data Frame Construction and Manipulation (BL4 – Analyze))

A dataset needs to be organized efficiently.

Write pseudocode to:

● Create a data frame with multiple columns (numeric and character)

● Add, delete, and modify columns

● Perform basic summary operations

Pseudocode:

BEGIN

CREATE a data frame named Student

ADD columns:

ID

Name

Age

Department

Marks

DISPLAY the data frame

ADD a new column called Grade

CALCULATE Grade based on Marks

MODIFY the Marks column

DELETE the Age column

CALCULATE:

Average Marks

Maximum Marks

Minimum Mark

DISPLAY the updated data frame

DISPLAY the summary results

END

5. Functions and Logical Operations (BL4 – Analyze)

Design pseudocode to implement functions and logical vector operations:

● Create a user-defined function with input parameters

● Perform operations on logical vectors

● Return and display computed results

Pseudocode:

BEGIN

DEFINE a function named calculateAverage

ACCEPT three marks as input

CALCULATE total marks

```
    CALCULATE average marks
    RETURN average marks
  END FUNCTION
  CALL calculateAverage with suitable marks
  STORE returned value in average
  DISPLAY average
  CREATE a vector marks
  CREATE a logical vector by checking:
    marks > 50
  DISPLAY logical vector
  SELECT marks where logical value is TRUE
  DISPLAY selected marks
END
```

Code of Conduct:

I Dhanushree S/192421218 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dhanushree S

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, and constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient